

Girish Yandigeri

PROGRAM – 10 (NGINX Deployment & Service in Kubernetes)

Commands

`minikube start --cpus=2 --memory=4096`
Starts Minikube with 2 CPUs and 4GB RAM.

`minikube dashboard`
Opens the Kubernetes Dashboard.

`kubectl create deployment nginx-deployment --image=nginx --replicas=2`
Creates an NGINX deployment with 2 replicas.

`kubectl get deployments`
Displays all deployments.

`kubectl get pods`
Shows the pods created for the deployment.

`kubectl expose deployment nginx-deployment --type=NodePort --port=80`
Creates a NodePort service to expose the deployment on port 80.

`kubectl get services`
Lists all services in the cluster.

`kubectl describe service nginx-deployment`
Shows detailed info about the exposed service.

`kubectl get pods --show-labels`
Displays pod labels.

`kubectl patch service nginx-deployment -p '{"spec":{"selector":{"app":"nginx-deployment"}}}'`
Fixes the service selector to match the deployment label.

`kubectl describe service nginx-deployment`
Verifies updated service details.

`kubectl get endpoints nginx-deployment`
Shows the pod IPs linked to the service.

`minikube service nginx-deployment`
Opens the NGINX page in the browser using the NodePort URL.

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab10
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl create deployment nginx-deployment --image=nginx --replicas=2
deployment.apps/nginx-deployment created
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl get deployments
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment                    0/2      2             0            9s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-7457467ffd-vzgtl   0/1     ContainerCreating   0           9s
nginx-deployment-7457467ffd-xlsvs   0/1     ContainerCreating   0           9s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-7457467ffd-vzgtl   1/1     Running             0          25s
nginx-deployment-7457467ffd-xlsvs   1/1     Running             0          25s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl get services
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes  ClusterIP   10.96.0.1     <none>        443/TCP    7d9h
nginx-deployment  NodePort    10.105.100.67 <none>        80:32358/TCP  6s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ kubectl describe service nginx-deployment
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointsSlice
Name:      nginx-deployment
Namespace: default
Labels:    app=nginx-deployment
Annotations: <none>
Selector:  app=nginx-deployment
Type:      NodePort
IP Family Policy: SingleStack
IP Families: IPv4
IP:         10.105.100.67
IPs:        10.105.100.67
Port:      <unset> 80/TCP
TargetPort: 80/TCP
NodePort:   <unset> 32358/TCP
Endpoints:  10.244.0.35:80,10.244.0.36:80
Session Affinity: None
External Traffic Policy: Cluster
Events:     <none>
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ minikube service nginx-deployment
NAMESPACE  NAME          TARGET PORT  URL
default    nginx-deployment  80           http://192.168.49.2:32358
Opening service default/nginx-deployment in default browser...
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$ Opening in existing browser session.
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/normallabs/lab10$
```

minikube dashboard

Confirms pod and service status in dashboard.

kubectl delete service nginx-deployment

Deletes the NodePort service.

kubectl delete deployment nginx-deployment

Deletes the NGINX deployment.

kubectl get deployments

Checks whether deployments still exist.

kubectl get pods

Checks whether pods still exist.

kubectl get services

Checks whether the service still exists.